

HYDROGENATED NBR: POLYMERIZATION, DEGRADATION, HYDROGENATION, AND WORK-UP STRATEGIES FROM PATENT EVIDENCE

1. ABSTRACT

The patent set spans the full hydrogenated nitrile rubber (HNBR) value chain, from NBR polymerization and latex work-up to post-polymer hydrogenation and downstream compounding. The core target is hydrogenated NBR, defined here as nitrile rubber in which the butadiene-derived carbon-carbon double bonds are partially or fully saturated while retaining nitrile functionality. Across the evidence, the dominant technical objective is to improve heat resistance, oil resistance, ozone resistance, and processability while controlling molecular weight, Mooney viscosity, residual unsaturation, and storage stability. The older patents emphasize metathesis-based molecular-weight reduction of NBR prior to hydrogenation, using ruthenium- or osmium-based catalysts, sometimes with co-olefins, to obtain lower-MW feedstocks that hydrogenate to lower-viscosity HNBR. These disclosures highlight the challenge that direct hydrogenation increases Mooney viscosity and that conventional mastication can damage microstructure. A later continuous hydrogenation patent shifts the focus to reactor engineering, using a multistage agitated reactor to improve heat and mass transfer for continuous hydrogenation of unsaturated polymers, including NBR-type substrates, with homogeneous or heterogeneous catalysts. Another patent introduces catalyst-system additives, specifically salts such as alkaline earth metal chlorides, to improve metathesis catalyst activity for NBR degradation, although copper salts are shown to be detrimental in this context. More recent disclosures address manufacturing and work-up: one patent uses a first coagulation of NBR latex with halogen-free metal salts, hydrogenation in solution, and a second coagulation with divalent metal salts to produce HNBR with improved heat-oil resistance and metal adhesion; another focuses on highly saturated nitrile rubber compositions blended with polyamide resin, with specified nitrile and dicarboxylic monoester contents and iodine value limits. The landscape also includes adjacent rubber-compounding patents that mention hydrogenated nitrile rubber as a compatible base rubber, but these are formulation-oriented rather than synthesis-oriented. Overall, the evidence supports a synthesis strategy combining controlled NBR composition, low-damage molecular-weight adjustment where needed, efficient solution hydrogenation, and carefully chosen coagulation/work-up conditions to preserve performance and processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10815320B2
- Patent Title: Functionalized resin having a polar linker
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US10837947B2
- Patent Title: Modified resins and uses thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: US8530584B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 7

- Patent Number: EP3708590B1
- Patent Title: Method for manufacturing hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: EP2147933B1
- Patent Title: Process for the continuous hydrogenation of carbon-carbon double bonds in an ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US7923518B2
- Patent Title: Nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 10

- Patent Number: EP2692788B1
- Patent Title: Highly saturated nitrile rubber composition and crosslinked rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples Comparative Example 1: - Example Extracted Examples Comparative Example 2: - Example Extracted Examples Comparative Example 1: - Example Extracted Examples Comparative Example 3: - Example Extracted Examples Comparative Example 2: - Example Extracted Examples Comparative Example 4: - Example Extracted Examples Comparative Example 5: - Example Extracted Examples Comparative Example 6: - Example Extracted Examples Comparative Example 6: - Example Extracted Examples Comparative Example 7: - Example Extracted Examples Comparative Example 7: - Example Extracted Examples Comparative Example 8: - Example Extracted Examples Comparative Example 4: - Example Extracted Examples Comparative Example 8: - Example Extracted Examples Synthesis Example 1: - Example Extracted Examples Synthesis Example 2: - Example Extracted Examples Synthesis Example 3: - Example Extracted Examples Synthesis Example 1: - Example Extracted Examples Example 1: - Example Extracted Examples Synthesis Example 1:

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2 and US8530584B2 both use metathesis-based degradation of nitrile rubber before or in conjunction with hydrogenation, but US7919563B2 explicitly targets low molecular weight HNBR with MW 30,000 to 250,000, Mooney 3 to 50, and PDI < 2.5, while US8530584B2 focuses on catalyst-system enhancement using salts to improve metathesis activity and reduce catalyst loading.
- US7919563B2 specifies nitrile rubber containing at least one conjugated diene and at least one α,β -unsaturated nitrile, with optional comonomers, and uses Ru/Os metathesis catalysts in the presence of a co-olefin and solvent; US8530584B2 also uses Ru/Os-based metathesis catalysts but adds salts of general formula K_nA_Z- , with alkaline earth metal chlorides highlighted as beneficial.
- US7919563B2 emphasizes a two-step sequence: metathesis degradation followed by hydrogenation, preferably in situ, whereas US8530584B2 is directed to the metathesis stage itself and reports that copper salts accelerate the initial rate but worsen final degradation efficiency.
- EP2147933B1 differs from the metathesis patents by focusing on continuous hydrogenation of unsaturated polymers in a multistage agitated reactor; it does not claim a specific NBR composition range, but it does specify a conjugated diolefin-based polymer with at least one other copolymerizable monomer, solvent, catalyst, and continuous feed/withdrawal operation.
- EP2147933B1 allows either homogeneous or heterogeneous catalysts, with preferred homogeneous organometallic catalysts including rhodium, ruthenium, osmium, or iridium complexes, while the metathesis patents center on ruthenium/osmium carbene-type catalysts for chain scission prior to hydrogenation.
- EP3708590B1 is a work-up/process patent rather than a catalyst patent: it uses a first coagulation of NBR latex with a halogen atom-free metal salt, solution hydrogenation, and a second coagulation with a divalent metal salt, contrasting with the older metathesis-first route.
- EP3708590B1 specifies an iodine value of 120 or less for the final hydrogenated nitrile rubber and highlights magnesium sulfate or aluminum sulfate in the first coagulation and calcium chloride or magnesium chloride in the second coagulation; the older metathesis patents do not disclose such coagulation sequencing.
- US7923518B2 and EP2692788B1 are composition patents that include hydrogenated nitrile rubber as a component or base material, but they do not teach HNBR synthesis conditions; US7923518B2 emphasizes storage stability, calcium/chlorine content, and thio end groups in nitrile rubber, while EP2692788B1 defines highly saturated nitrile rubbers with nitrile and dicarboxylic monoester units blended with polyamide resin.
- US10731020B2, US10815320B2, and US10837947B2 are rubber-compounding or resin-modification patents that mention hydrogenated nitrile rubber only as one possible base polymer or adjacent elastomer class; they are not primary HNBR synthesis disclosures and provide no direct hydrogenation or NBR polymerization parameters.

- Across the primary synthesis patents, the recurring technical themes are: controlling unsaturation before hydrogenation, minimizing catalyst burden and reaction time, preserving microstructure, and improving downstream processability through reactor design or coagulation/work-up control.
- Historically, the route progresses from metathesis-based molecular-weight reduction of NBR followed by hydrogenation to more process-engineered continuous hydrogenation and improved latex coagulation/work-up, indicating a shift from chemistry-first to process-first optimization.

4. CONCLUSION

The patent evidence indicates that the most effective HNBR synthesis strategies combine controlled NBR feedstock design, selective hydrogenation, and process steps that preserve polymer integrity while improving processability. For low-viscosity HNBR, the strongest evidence supports pre-hydrogenation molecular-weight reduction by metathesis using ruthenium- or osmium-based catalysts, ideally with catalyst-activity enhancement by suitable salts and without copper additives. For direct hydrogenation manufacturing, continuous multistage agitated reactors offer a credible route to better heat and mass transfer and more stable operation. For latex-based manufacture, a two-stage coagulation sequence using a halogen-free metal salt before hydrogenation and a divalent metal salt after hydrogenation appears advantageous for obtaining HNBR with good heat-oil resistance and improved handling. Based on the disclosed evidence, recommended synthesis parameters are: NBR feedstocks with butadiene-derived unsaturation suitable for selective hydrogenation, hydrogenation to a low residual double-bond level consistent with the cited HNBR definitions, and work-up/coagulation conditions that minimize contamination and preserve storage stability. Where lower Mooney viscosity is required, metathesis degradation before hydrogenation is the clearest route; where manufacturing robustness is prioritized, continuous hydrogenation and optimized coagulation are the preferred process directions.

5. REFERENCES

- US10815320B2 | Functionalized resin having a polar linker | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10815320B2/en>
- US10837947B2 | Modified resins and uses thereof | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10837947B2/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US8530584B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8530584B2/en>

- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- EP3708590B1 | Method for manufacturing hydrogenated nitrile rubber | Not disclosed | EP | 2022 | <https://patents.google.com/patent/EP3708590B1/en>
- EP2147933B1 | Process for the continuous hydrogenation of carbon-carbon double bonds in an ... | Not disclosed | EP | 2012 | <https://patents.google.com/patent/EP2147933B1/en>
- US7923518B2 | Nitrile rubbers | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7923518B2/en>
- EP2692788B1 | Highly saturated nitrile rubber composition and crosslinked rubber | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2692788B1/en>